

SEMESTER 2 EXAMINATION 2025/26

ADVANCED MACHINE LEARNING

DURATION 120 MINS (2 HOURS)

This paper contains FOUR questions.

Student ID:

Answer all parts of the question in section A (40 marks) and TWO questions from section B (30 marks each)

An outline marking scheme is shown in square brackets to the right of each question.

You **must** write your Student ID in the box and must not write your name anywhere on the paper.

Your answers should be written in the boxes provided on the question paper. The “Additional Space” provided on the facing or adjacent page to the question may be used for corrections or extra space. Clearly indicate which question the answer corresponds to.

Scrap paper can be used - this will be discarded without being looked at.

University approved calculators MAY be used.

FOR EXAMINER USE ONLY

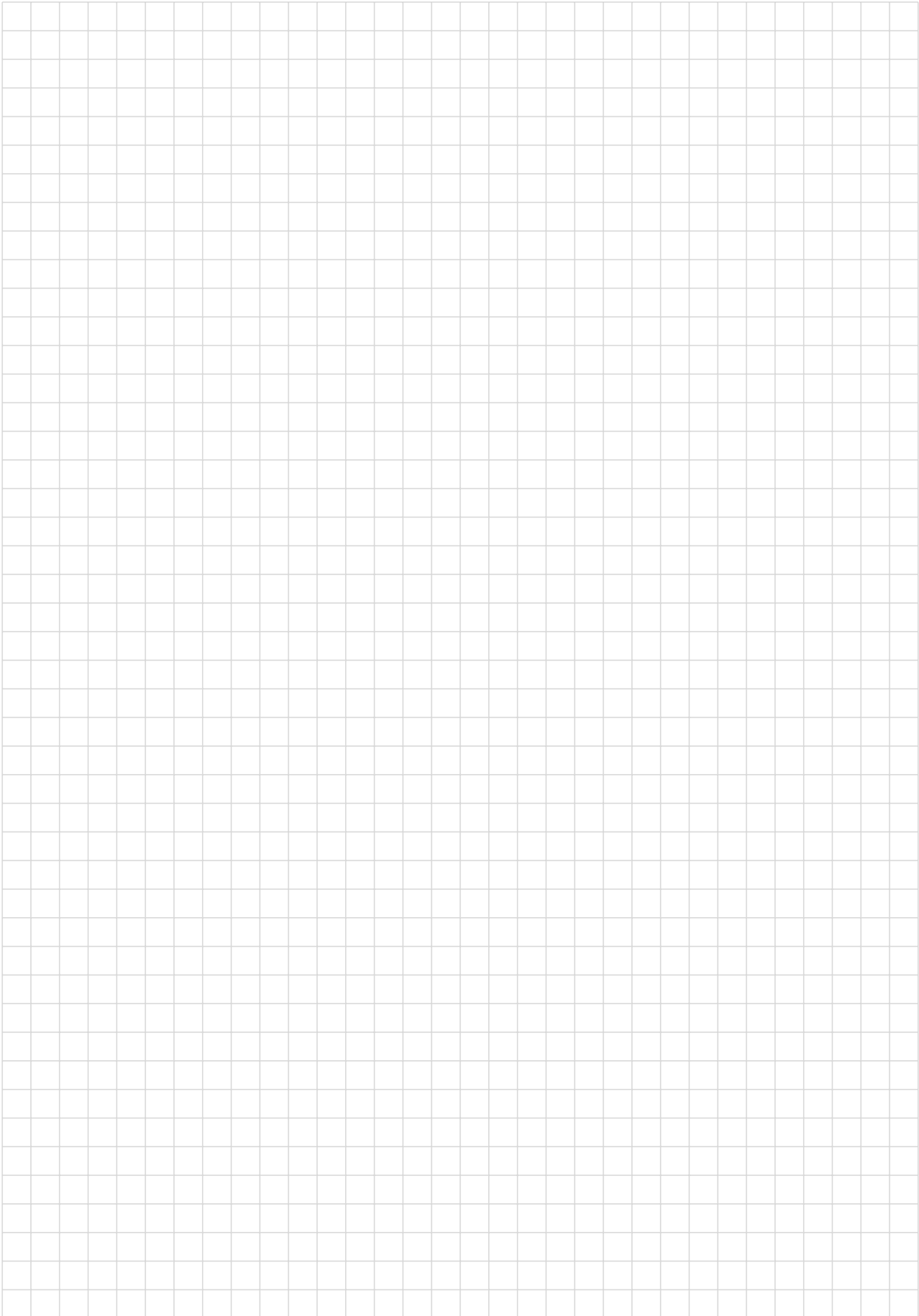
Question	Mark	Arithmetic checked	Double Marked
A1	/40		
B2	/30		
B3	/30		
B4	/30		
Total:			

A foreign language translation dictionary (paper version) is permitted provided it contains no notes, additions or annotations.

This examination does not contain colour figures.

34 page examination paper. This paper has no handouts.

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Section A

A1

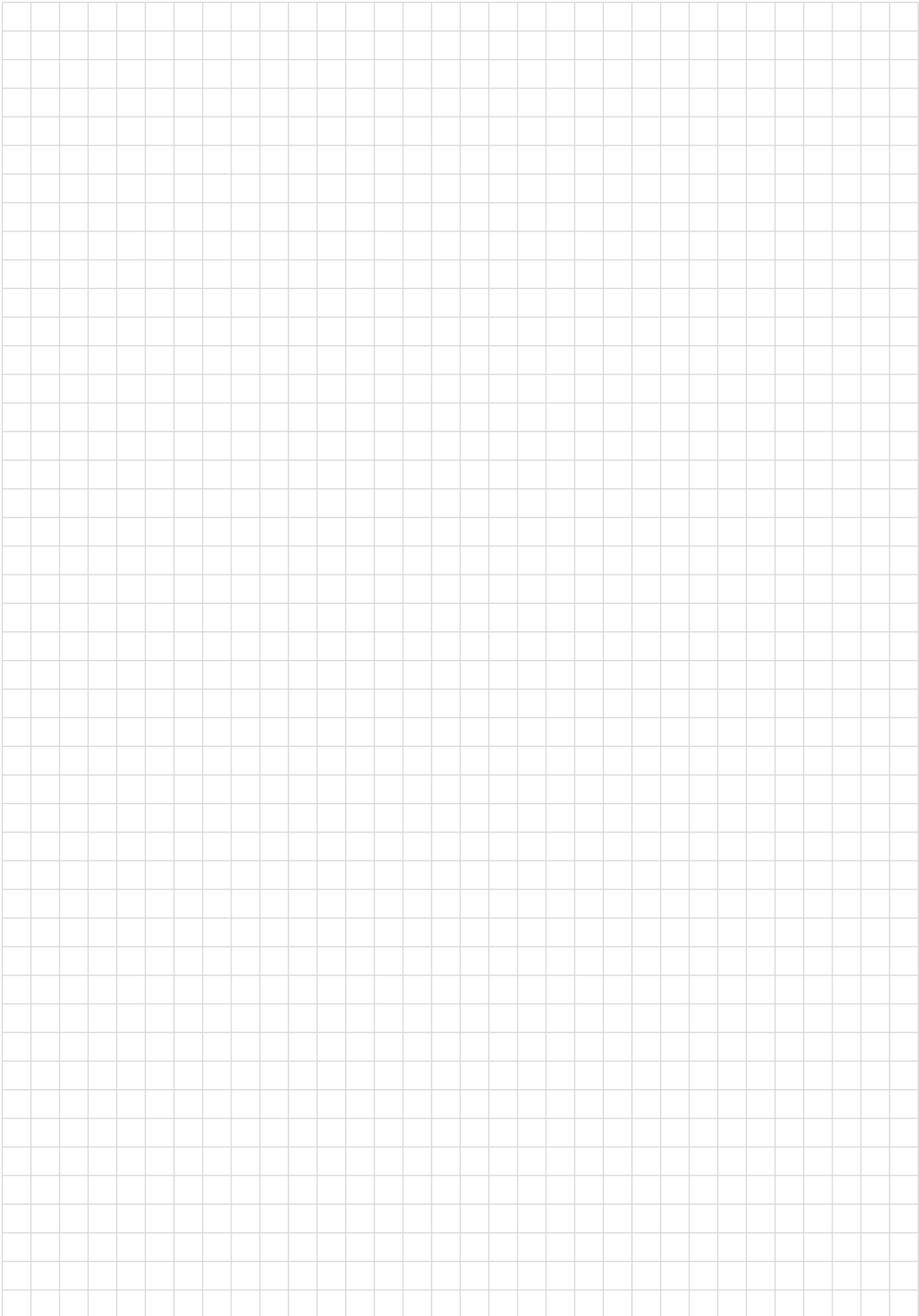
- (a) Explain what *data augmentation* is with a couple of examples of how it is done. Also provide an explanation of what its purpose is. [5 marks]

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- (b) Provide a brief overview of the *AdaBoost*, explain what task it solves, the structure of the classifier and how it is trained (exact formulae are not necessary). [5 marks]

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Additional space. Do not use unless necessary. Clearly mark corresponding question.



(c) Explain the kind of problems where the following learning machines excel

- (i) Random Forest
- (ii) Gradient Boosting
- (iii) Support Vector Machines
- (iv) Convolutional Neural Networks
- (v) Gaussian Processes

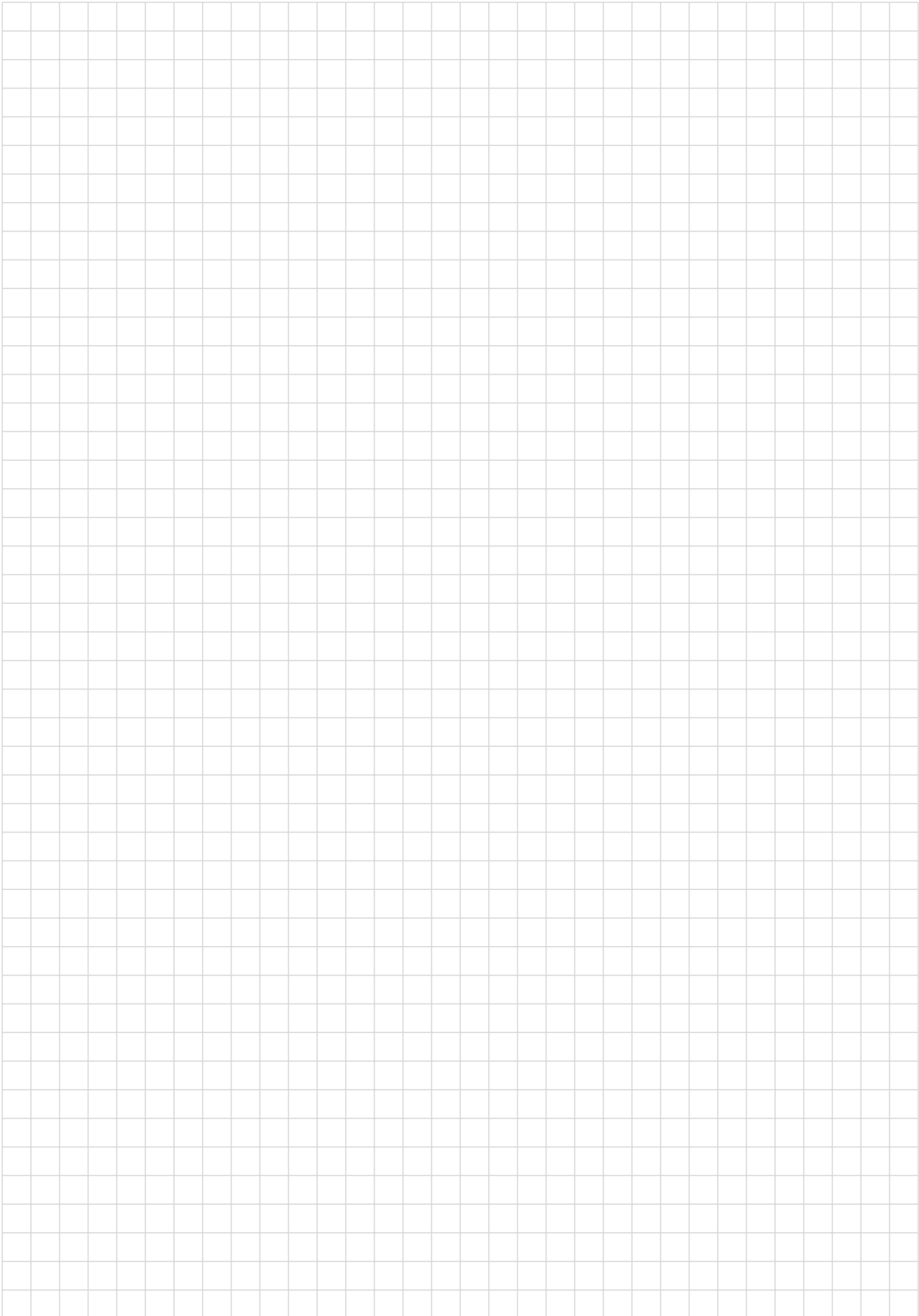
[5 marks]

i	
ii	
iii	
iv	
v	

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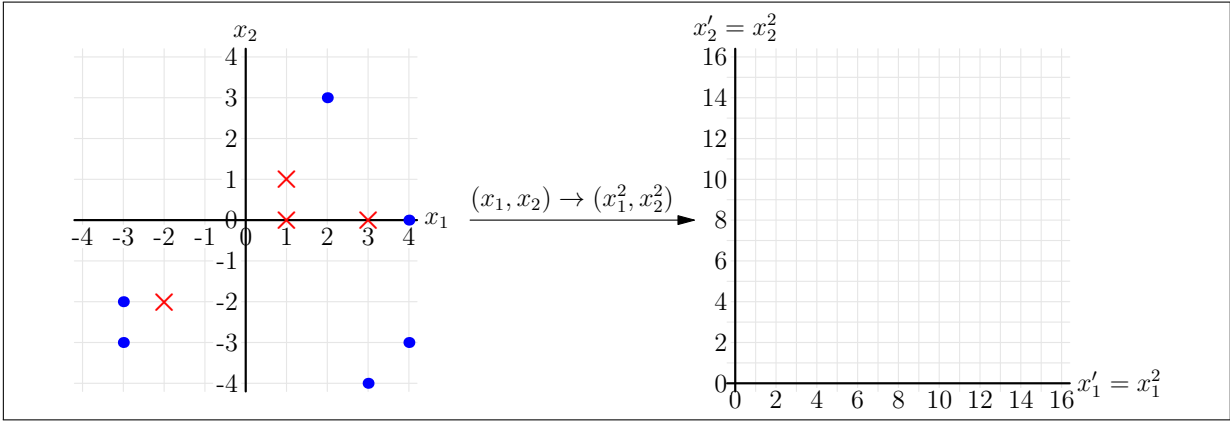
(d) Show that for the mapping

$$\mathbf{x} = (x_1, x_2, x_3) \rightarrow \vec{\phi}(\mathbf{x}) = (x_1^2, x_2^2, x_3^2, \sqrt{2} x_1 x_2, \sqrt{2} x_1 x_3, \sqrt{2} x_2 x_3)$$

the kernel $K(\mathbf{x}, \mathbf{y}) = \vec{\phi}(\mathbf{x}) \cdot \vec{\phi}(\mathbf{y})$ is equal to $(\mathbf{x} \cdot \mathbf{y})^2$. [5 marks]

5

(e) In the diagram below show how the datapoints gets projected under a projection $(x_1, x_2) \rightarrow (x_1^2, x_2^2)$. Sketch the rough position of the separating plane in the extended feature space. [5 marks]



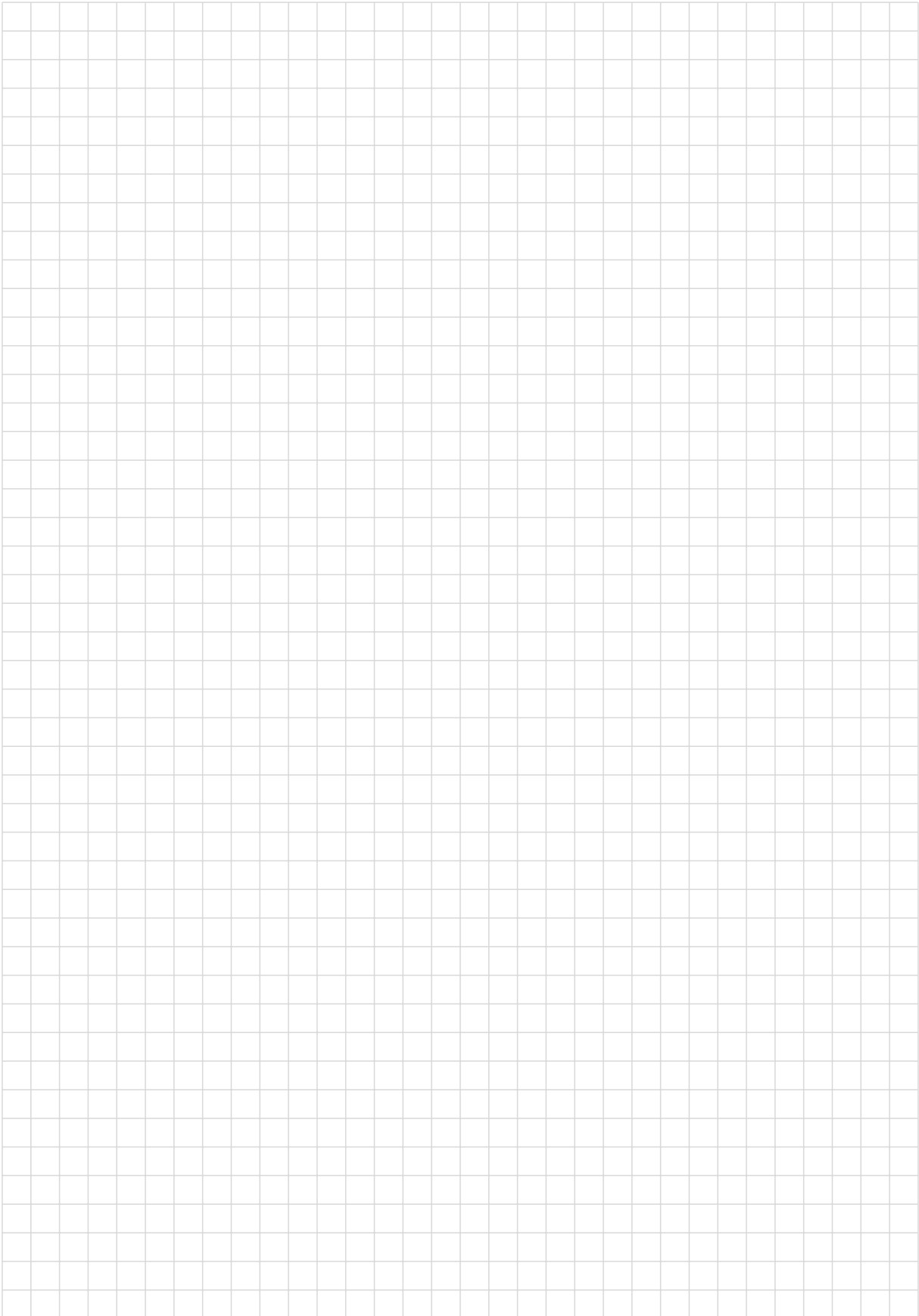
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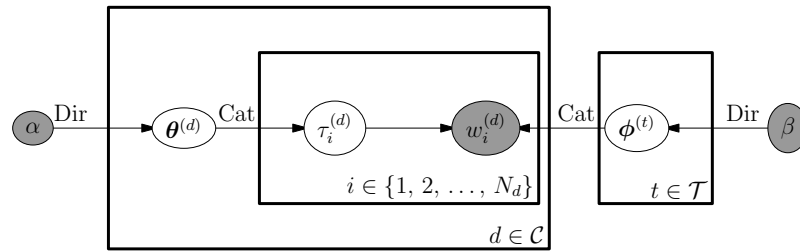
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- (f) Show that the beta distribution $\text{Beta}(p|a, b) = p^{a-1} (1 - p)^{b-1} / B(a, b)$ is a conjugate prior to the binomial likelihood $\text{Binom}(k|n, p) = \binom{n}{k} p^k (1 - p)^{n-k}$. Derive update equations for the parameters of the posterior distribution after observing k successes and $n - k$ failures.. [5 marks]

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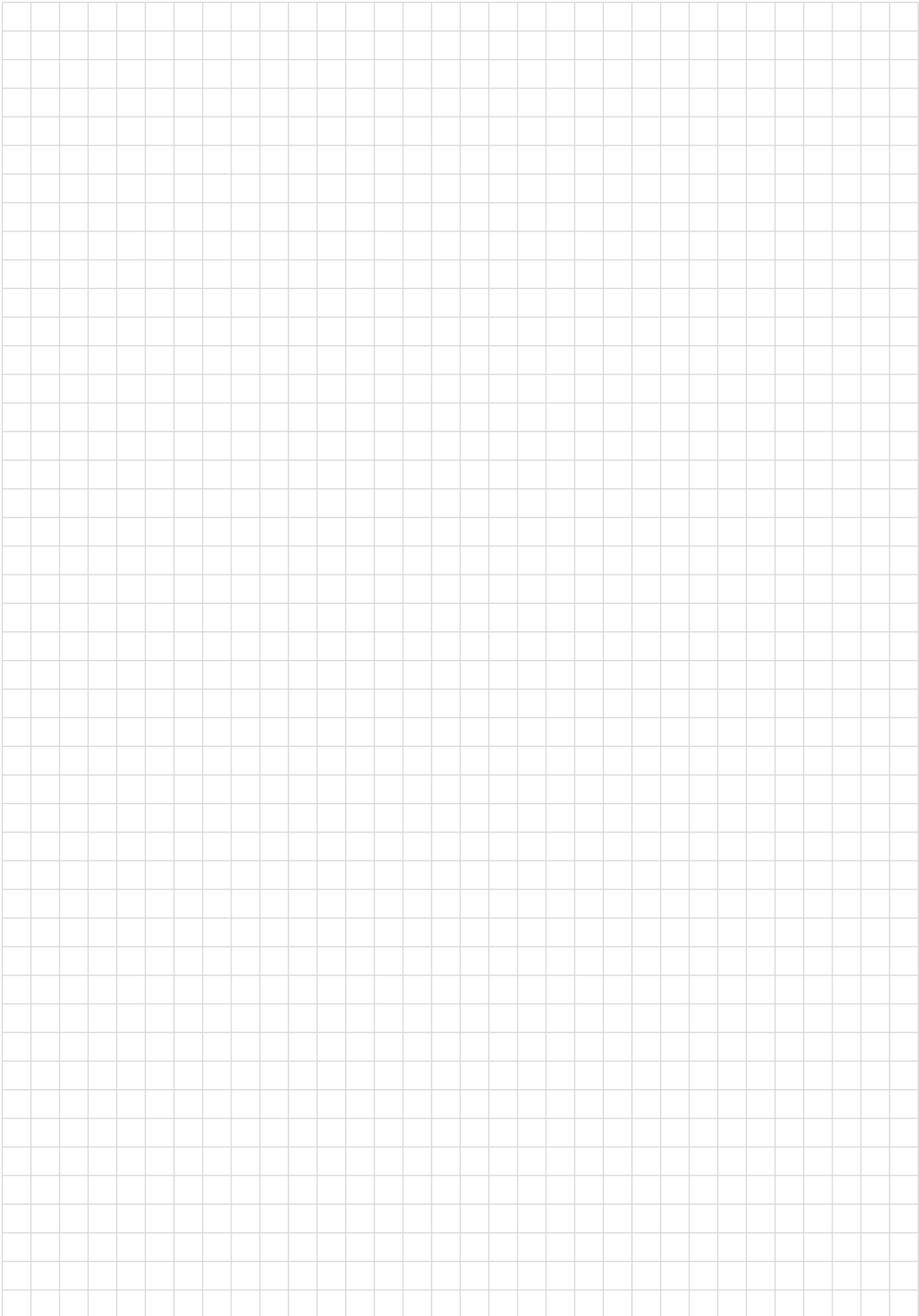
- (g) The smoothed latent Dirichlet allocation topic model can be represented as a graphical model by the following plate diagram



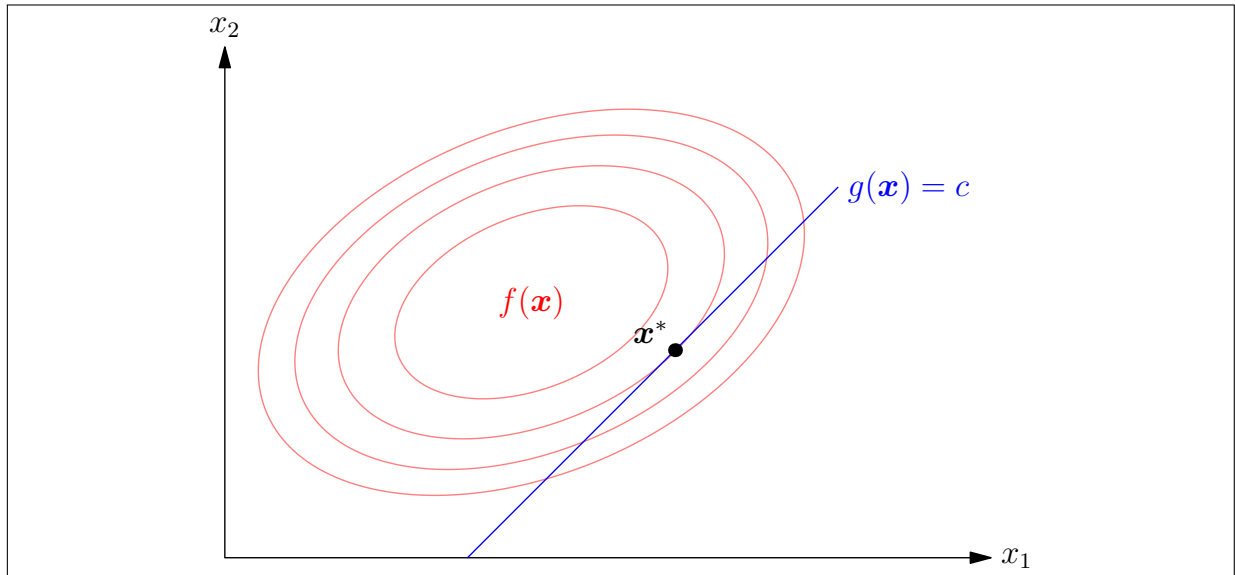
where $\mathcal{C}$ is a set of documents and $\mathcal{T}$ is the set of topics. Sketch how documents of size N_d are generated by expanding the plate diagram to show the full word generation process. [5 marks]

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- (h) Below is shown contour lines for a function $f(\mathbf{x}) = \mathbf{x}^T \mathbf{Q} \mathbf{x}$ that we wish to minimise subject to a linear constraint $g(\mathbf{x}) = c$. At the point $\mathbf{x}^*$ indicated, sketch the gradient $\nabla f(\mathbf{x}^*)$ and $\nabla g(\mathbf{x}^*)$. Use the diagram to explain why $\nabla \mathcal{L} = 0$ finds the solution to the constrained optimisation problem where $\mathcal{L} = f(\mathbf{x}) - \lambda g(\mathbf{x})$. [5 marks]

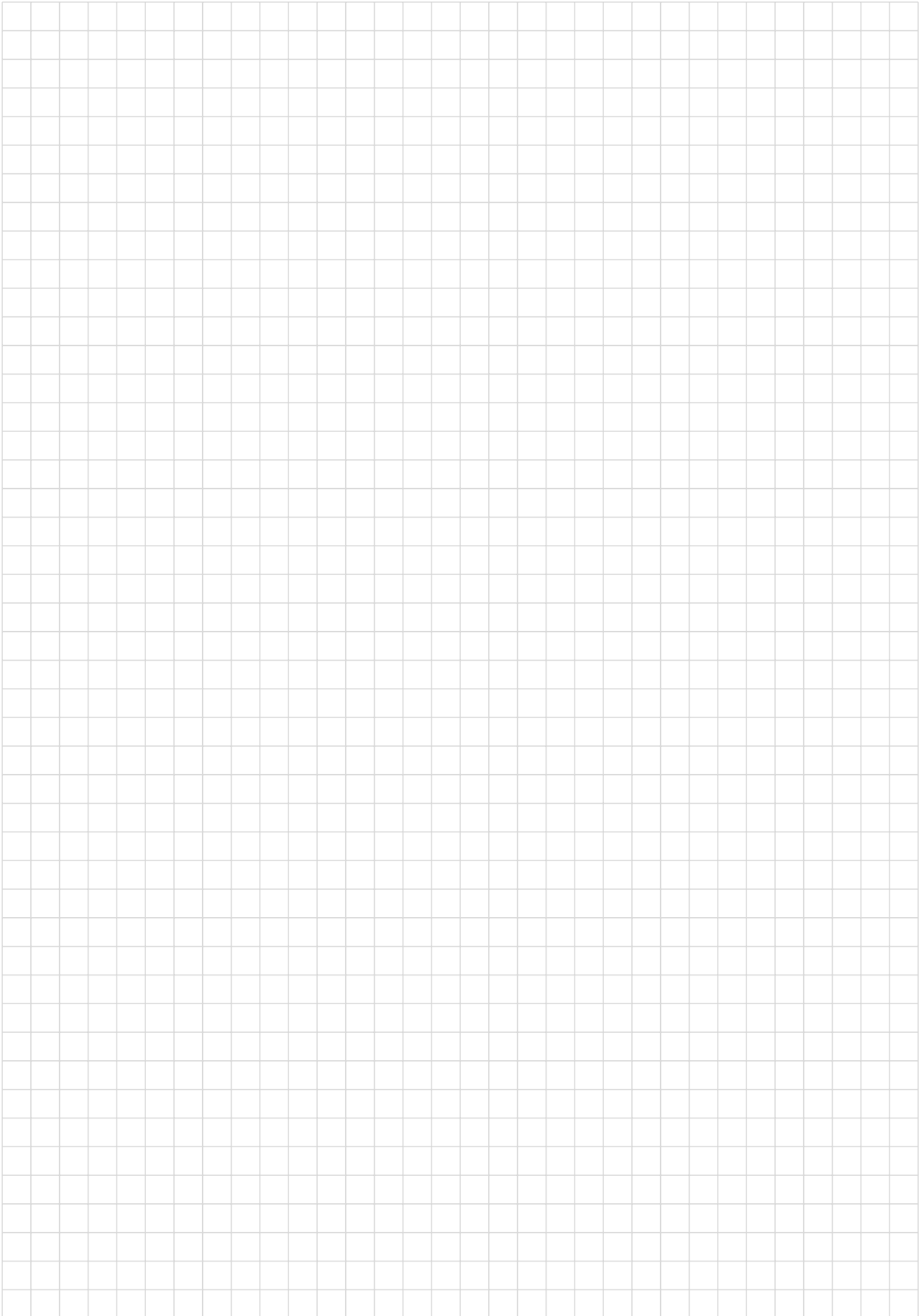


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End of question A1

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Section B

B2

(a) Show that the expected generalisation given by

$$\mathbb{E}_{\mathcal{D}}[E(\mathcal{D})] = \mathbb{E}_{\mathcal{D}} \left[\sum_{x \in \mathcal{X}} p(x) \left(\hat{f}(x|\mathcal{D}) - f(x) \right)^2 \right]$$

can be written as the sum of a bias term, B , and variance term V where

$$B = \sum_{\mathbf{x} \in \mathcal{X}} p(\mathbf{x}) \left(\hat{f}_m(\mathbf{x}) - f(\mathbf{x}) \right)^2, \quad V = \mathbb{E}_{\mathcal{D}} \left[\sum_{\mathbf{x} \in \mathcal{X}} p(\mathbf{x}) \left(\hat{f}(\mathbf{x}|\mathcal{D}) - \hat{f}_m(\mathbf{x}) \right)^2 \right]$$

where $\hat{f}_m(\mathbf{x}) = \mathbb{E}_{\mathcal{D}}[\hat{f}(\mathbf{x}|\mathcal{D})]$ is the prediction made by averaging over all machines. [1

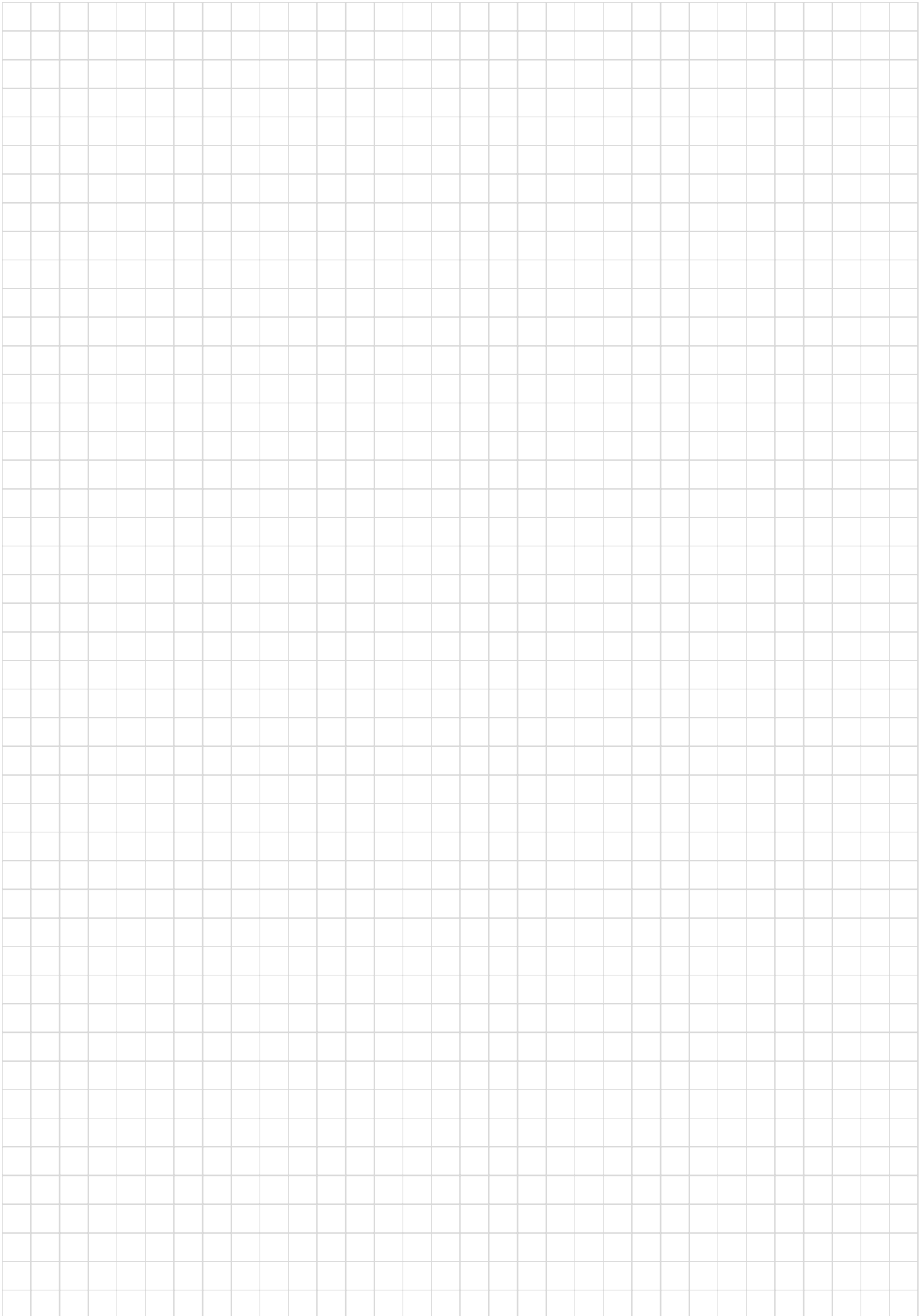
[15 marks]

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(b) Explain the *bias* and the *variance* terms in words.

[5 marks]

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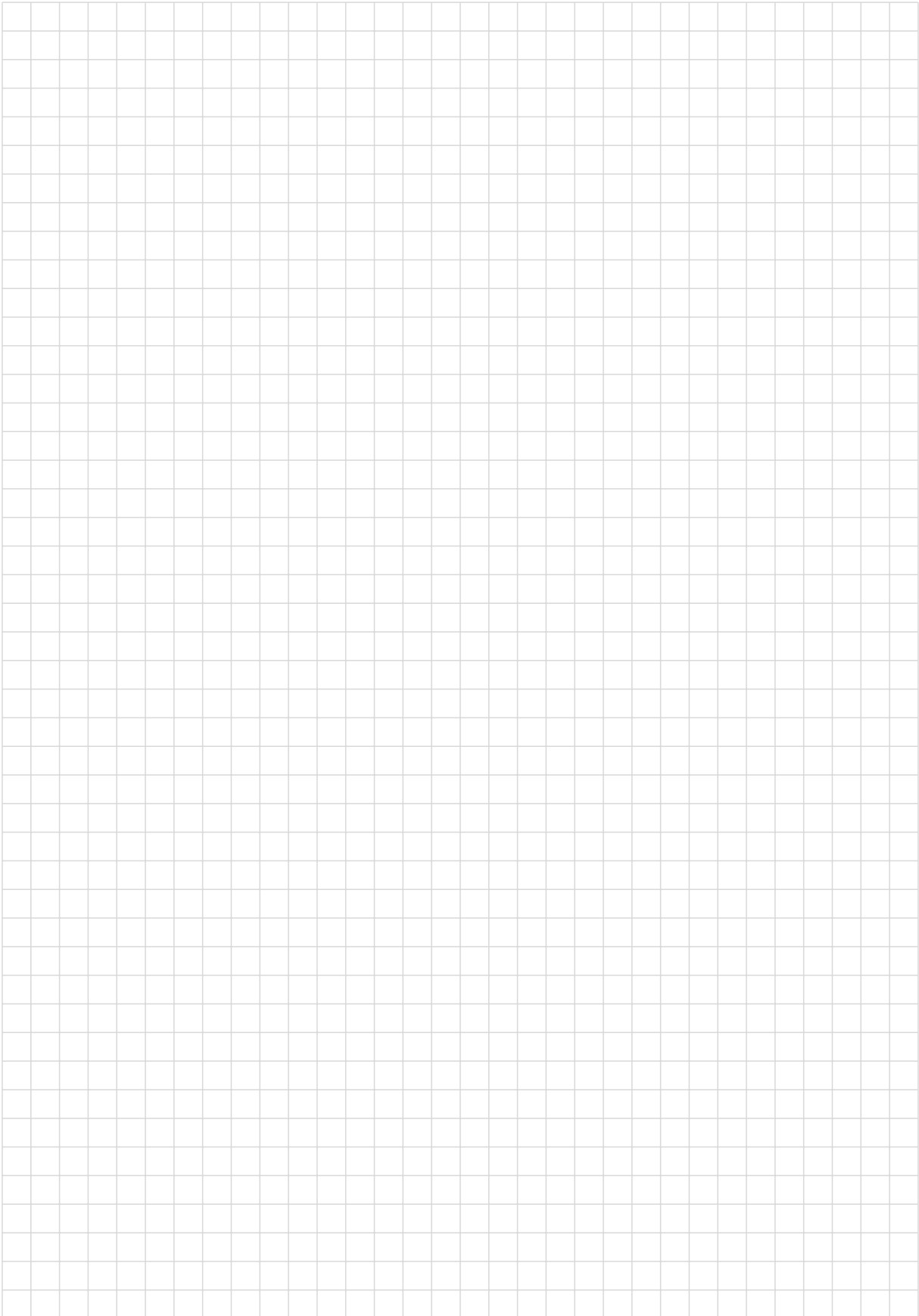
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(c) What is the bias-variance dilemma.

[5 marks]

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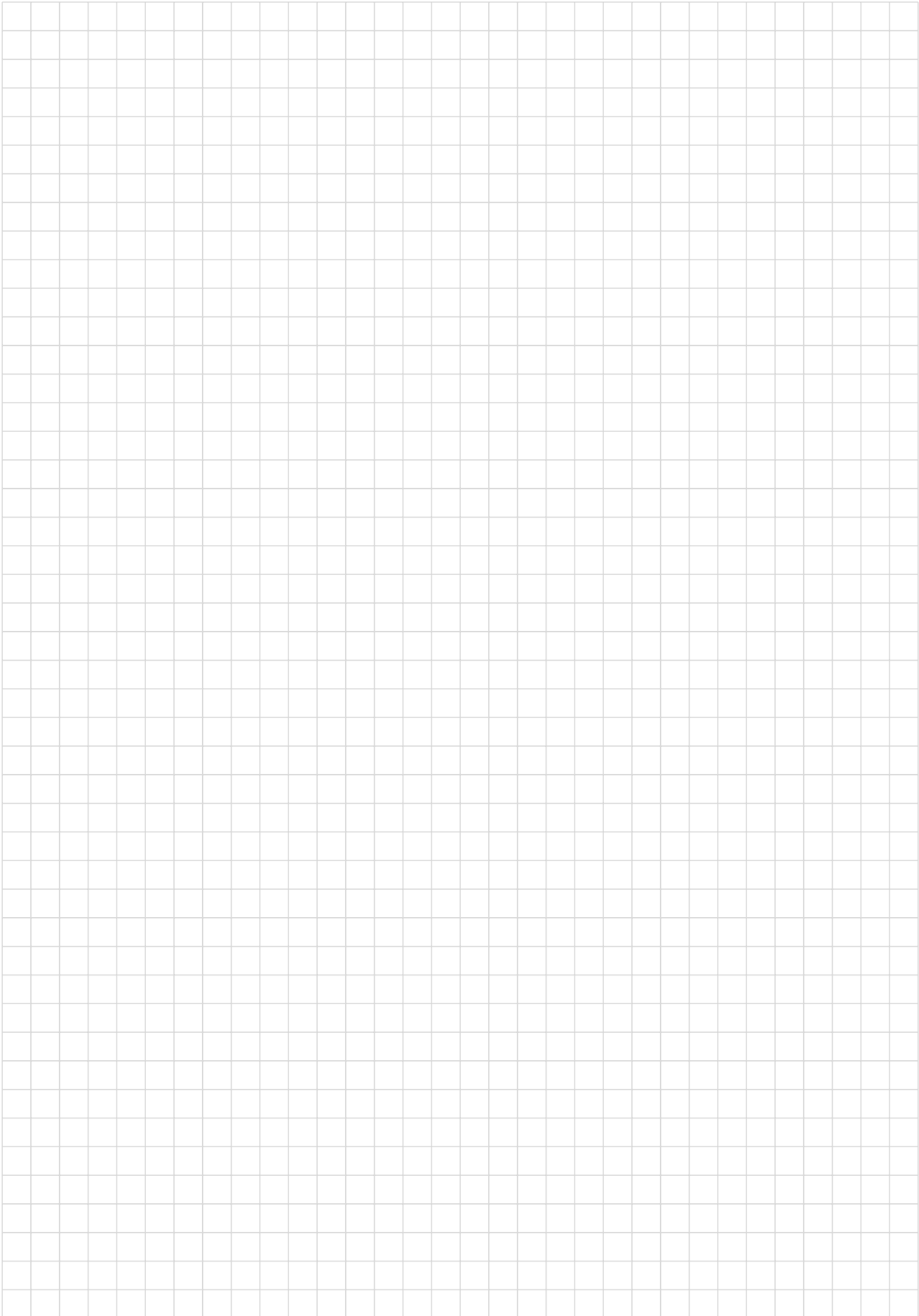
(d) Explain how regularisation helps in the context of the bias-variance dilemma. [5 marks]

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(a)	$\frac{\quad}{15}$	(b)	$\frac{\quad}{5}$	(c)	$\frac{\quad}{5}$	(d)	$\frac{\quad}{5}$	Total	$\frac{\quad}{30}$
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End of question B2

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B3

- (a) Explain for Gaussian Processes (GP) what is the prior, the likelihood and the posterior. [5 marks]

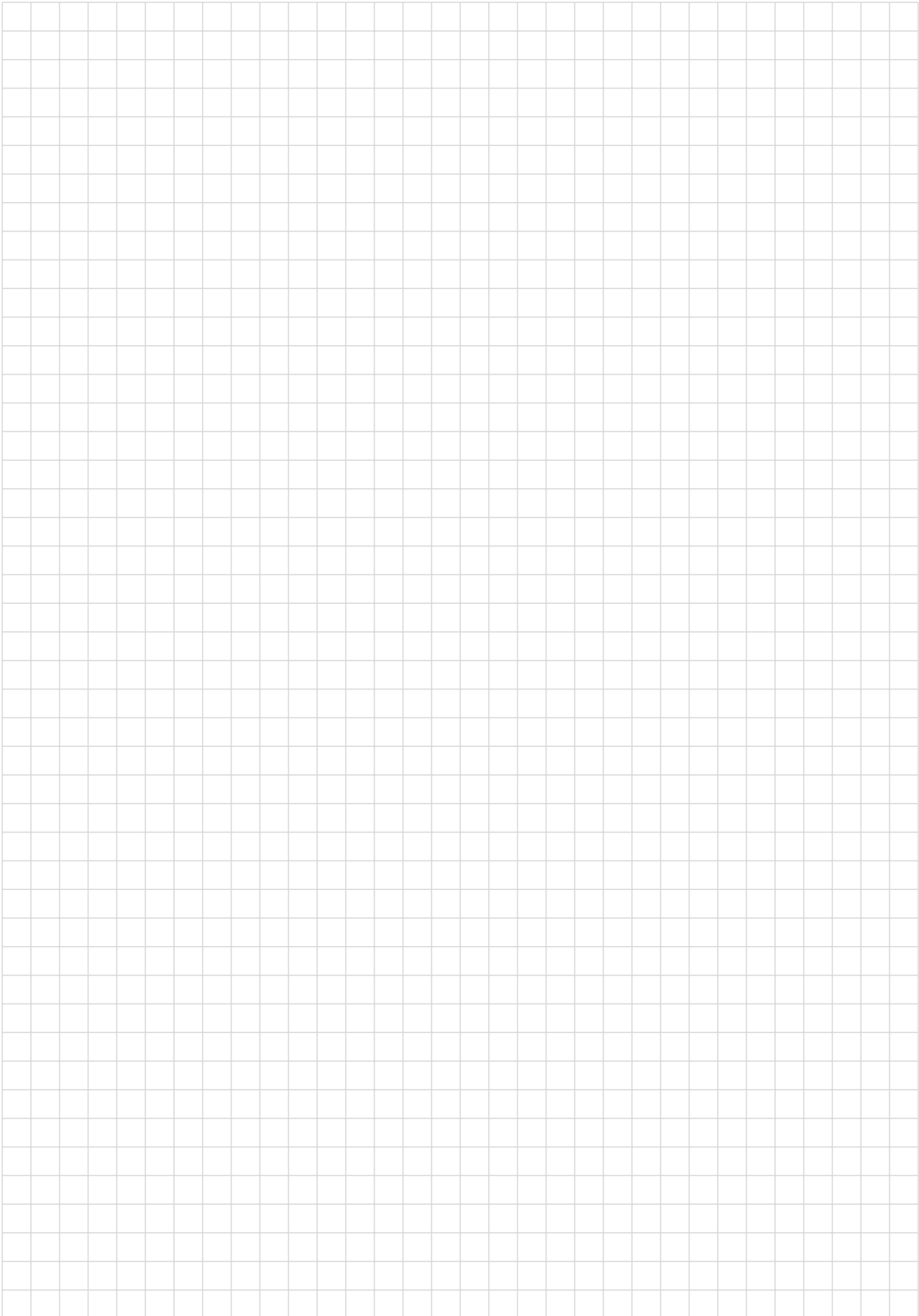
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- (b) Explain what the kernel function represents and how it could be measured empirically from many observations. [5 marks]

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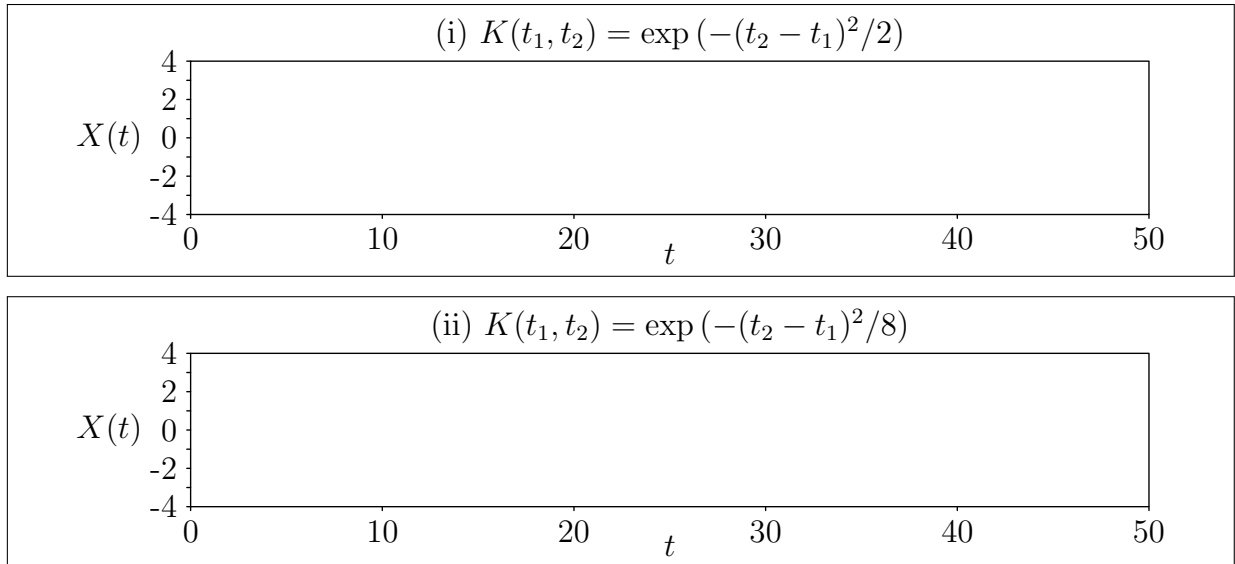
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(c) Consider a 1-d Gaussian Process, $X(t)$, with a kernel of the form

$$K(t_1, t_2) = \exp\left(-\frac{(t_2 - t_1)^2}{2\ell}\right).$$

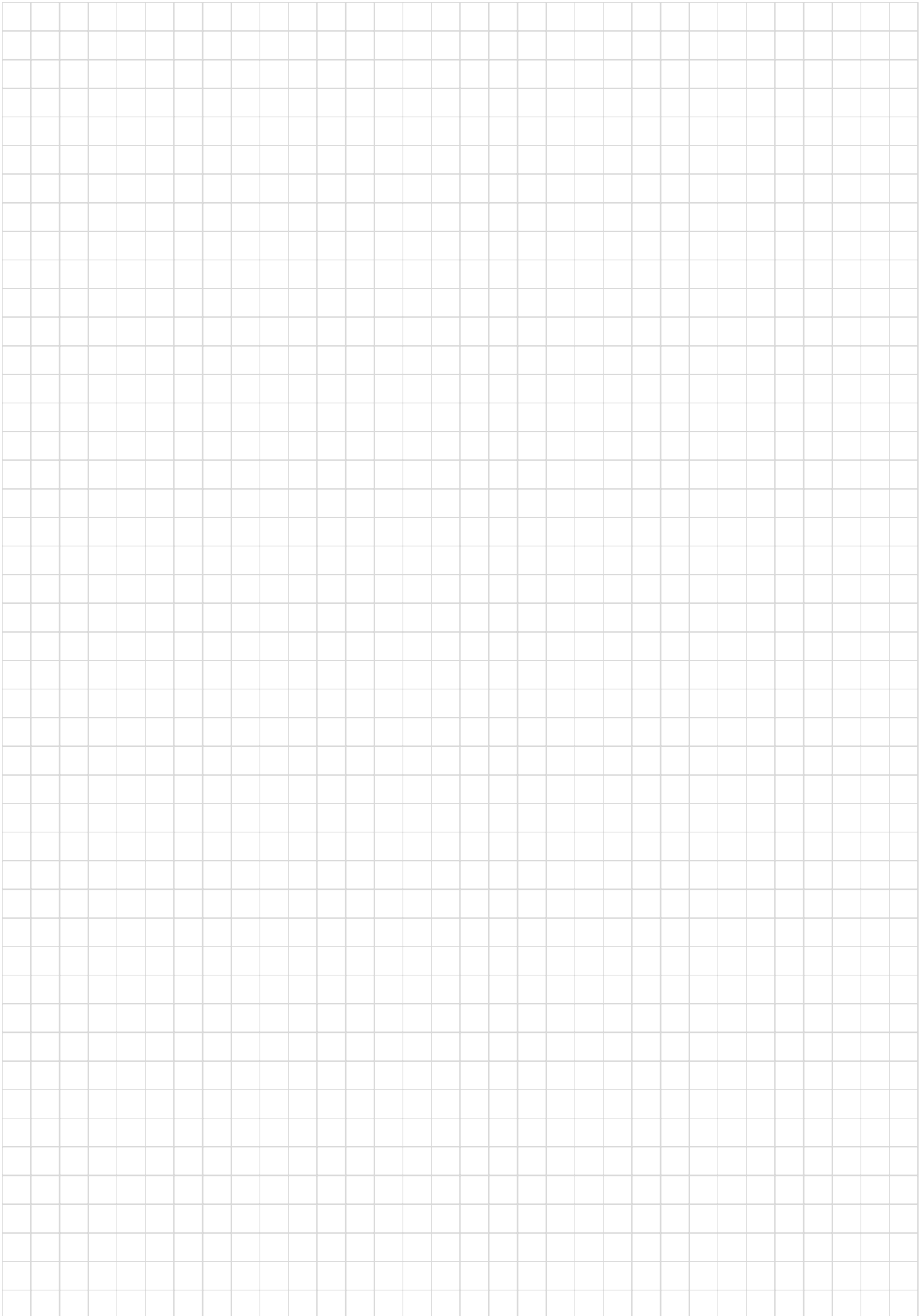
Sketch three Gaussian Processes drawn from the prior with (i) $\ell = 1$ and (ii) $\ell = 2$ (we are not looking for accuracy, but rather the effect of changing ℓ). [5 marks]



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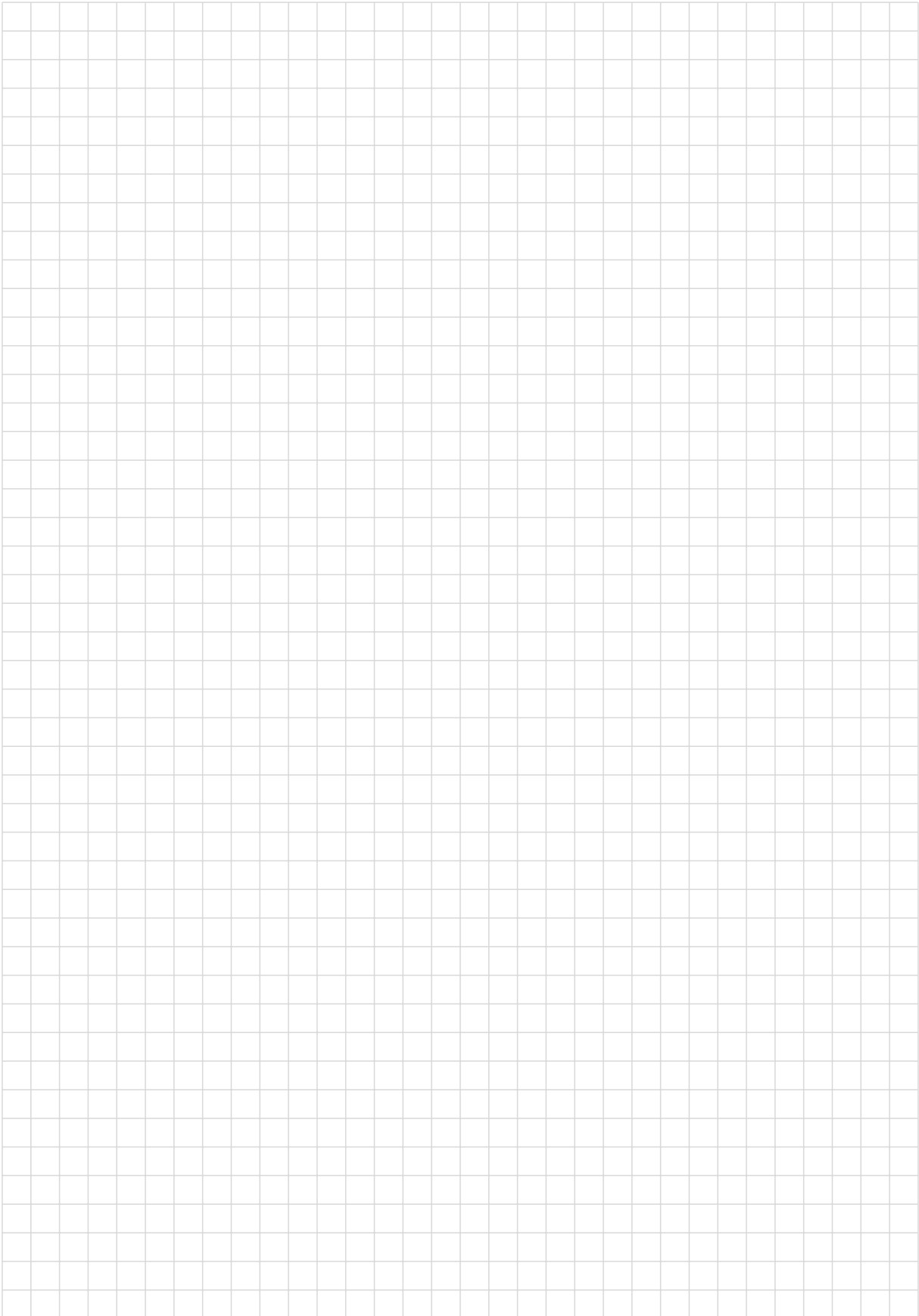


- (d) Explain the advantages and disadvantage of using the MAP solution rather than a full Bayesian solution. [5 marks]

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- (e) Explain why Monte Carlo techniques are often used to solve Bayesian inference problems. [5 marks]

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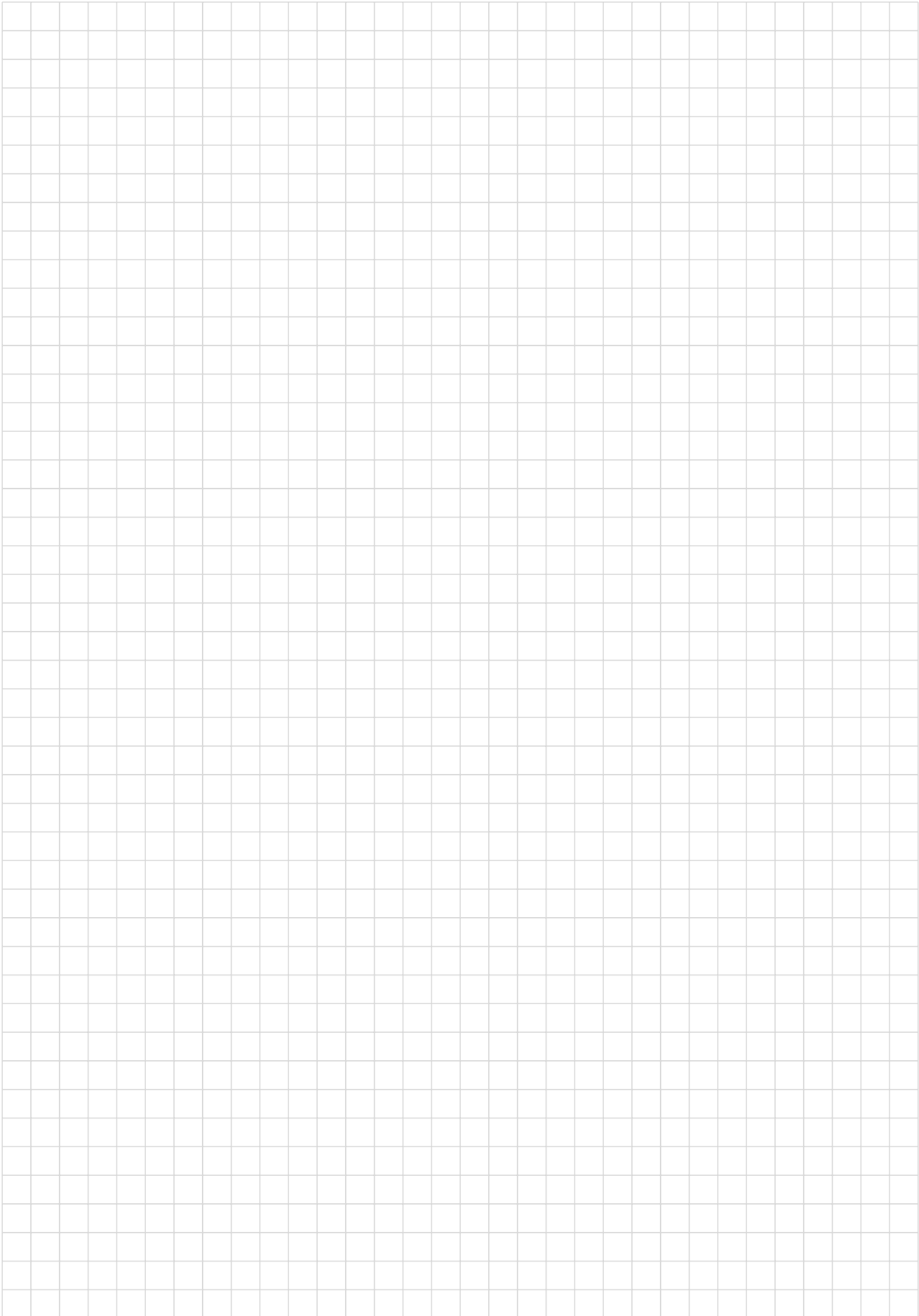
- (f) Briefly describe in words the use of the MCMC algorithm in Bayesian inference. [5 marks]

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End of question B3

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B4

- (a) What guarantees do we have when optimising a convex optimisation problem? [5 marks]

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- (b) By considering the Hessian of the loss

$$L(\mathbf{w}) = \frac{1}{2} \mathbf{w}^\top \mathbf{M} \mathbf{w}$$

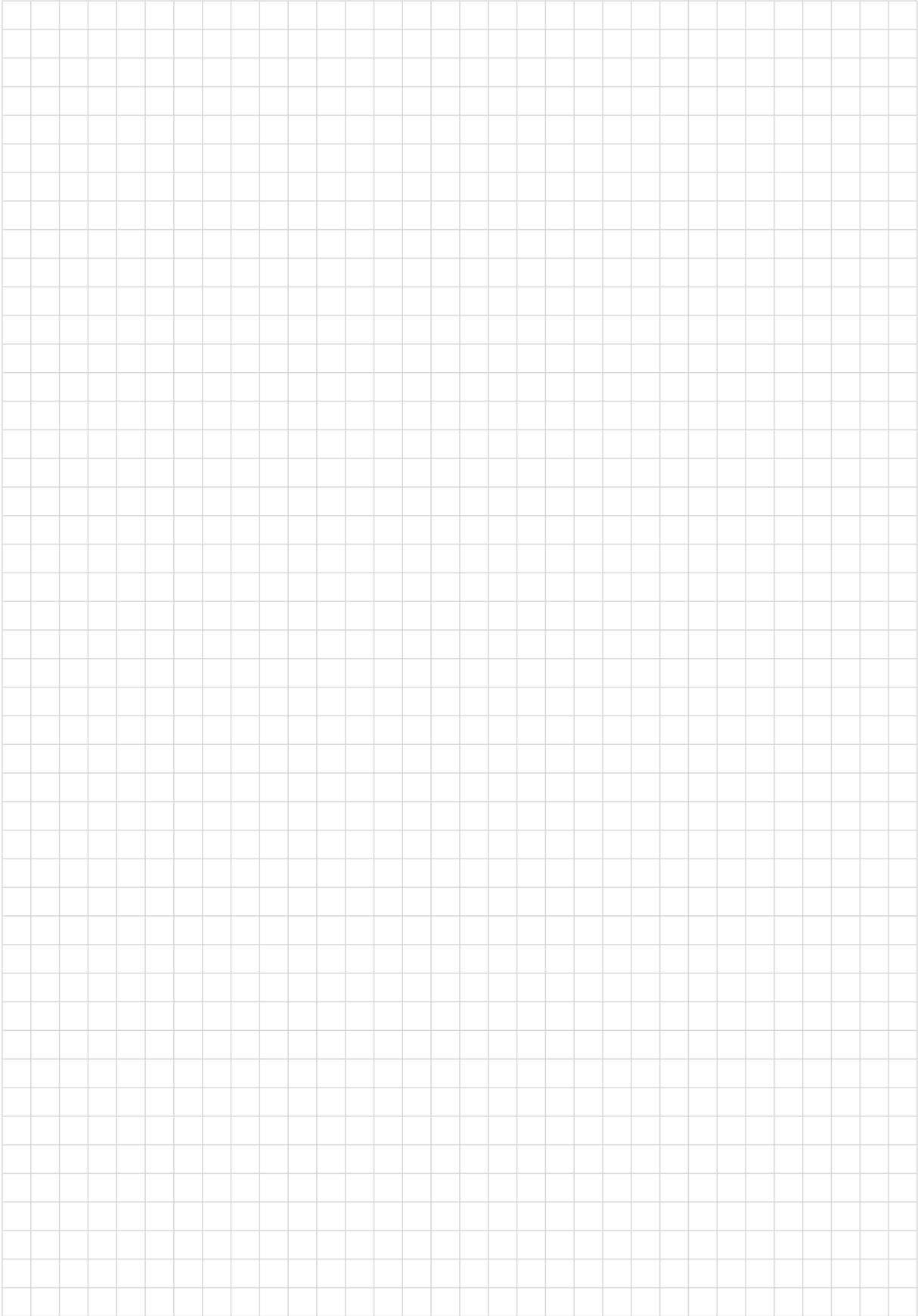
give the condition on the matrix $\mathbf{M}$ that must be satisfied for the loss function to be convex. [5 marks]

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- (c) By considering the second derivative show that $f(x) = -\log(x)$ is convex-up. [5 marks]

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(d) Jensen's inequality for convex-up function states that

$$\mathbb{E}[f(X)] \geq f(\mathbb{E}[X]).$$

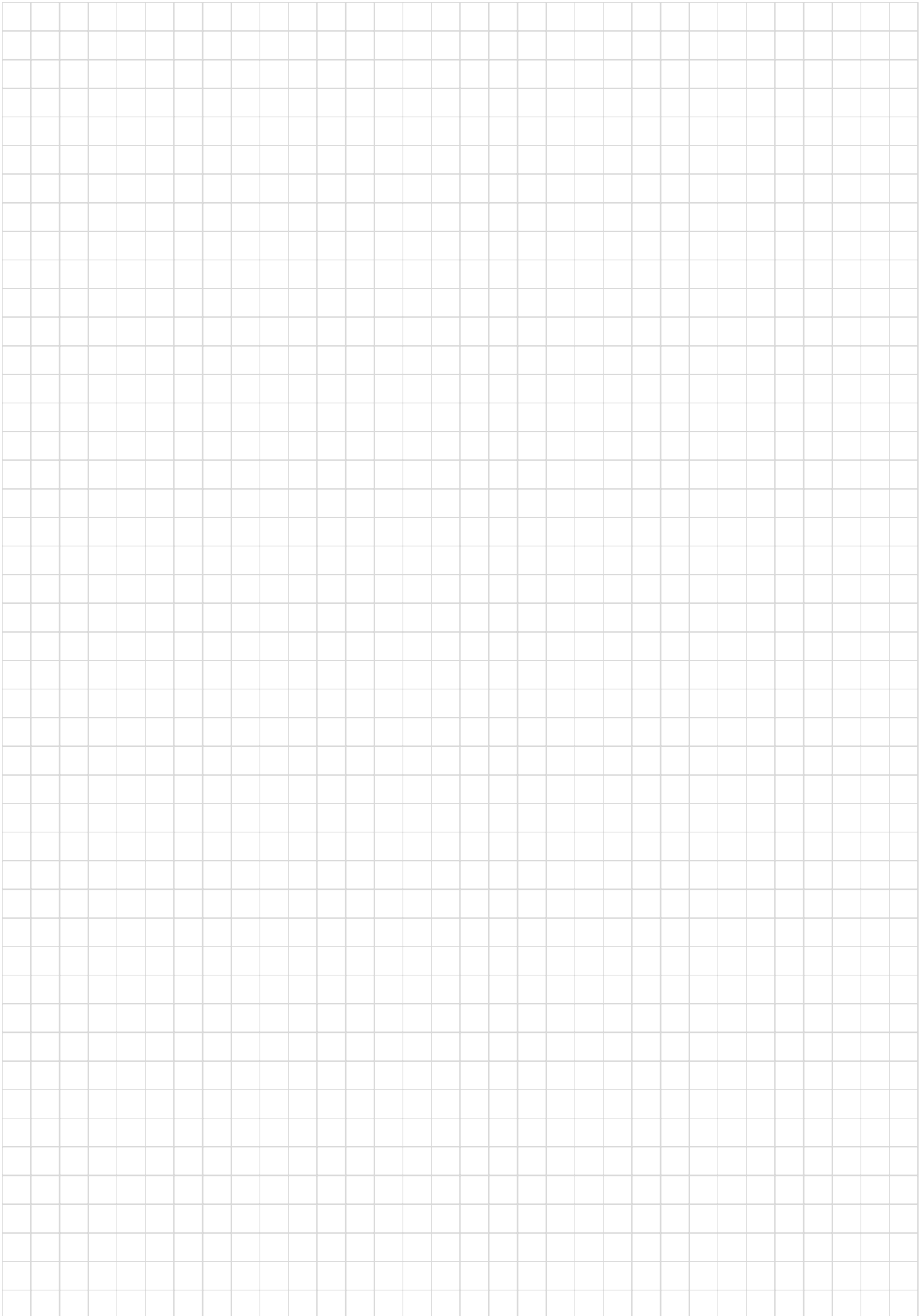
Use this to show that for any two categorical distributions $\mathbb{P}(F = i) = f_i$ and $\mathbb{P}(G = i) = g_i$, the KL-divergence

$$\text{KL}(F\|G) = -\sum_i f_i \log\left(\frac{g_i}{f_i}\right)$$

is non-negative.

[5 marks]

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- (e) In a variational auto-encoder an input image, x , is drawn from a dataset $\mathcal{D}$. The encoder generates a probability distribution $q_\phi(z|x)$ defined in a latent space. A vector in the latent space, z , is sampled from $q_\phi(z|x)$ and sent to the decoder that then generates a probability distribution in the space of images $p_\theta(x'|z)$. The loss function is defined as

$$\mathcal{L}(\phi, \theta) = \mathbb{E}_{x \sim \mathcal{D}} [\text{KL}(q_\phi(z|x) \parallel \mathcal{N}(\mathbf{0}, \mathbf{I})) - \log(p_\theta(x|z))]$$

Provide an interpretation of the two terms in the loss function.

[10 marks]

[illegible]

10

(a) $\frac{\quad}{5}$ (b) $\frac{\quad}{5}$ (c) $\frac{\quad}{5}$ (d) $\frac{\quad}{5}$ (e) $\frac{\quad}{10}$ Total $\frac{\quad}{30}$

End of question B4

END OF PAPER

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